

Question 1:

For an MP neuron with input size 3 and threshold condition

$$f(x) = 1 \text{ if } x_1 + x_2 + x_3 > 1$$

Which of the following input-output pairs are correct?

Options:

A) $x = (1,0,0), y = 0$

B) $x = (0,1,1), y = 1$

C) $x = (0,0,0), y = 0$

D) $x = (1,1,0), y = 1$

E) $x = (1,1,1), y = 0$

Correct Answers : A, B, C, D

Data for the following 3 questions (2, 3, 4)

A smart home system uses three binary sensors:

x_1 : Door sensor (1 if open)

x_2 : Motion detector (1 if movement)

x_3 : Window sensor (1 if open)

The MP neuron outputs 1 only when the total number of activated sensors exceeds 2. This output triggers an alert to the homeowner.

Question 2:

What is the minimum number of input values that must be 1 for the MP neuron to output 1?

Fill in the Textbox (type answer) _____

Correct Answer : 3

Explanation : Because the sum must be greater than 2, i.e., at least 3 inputs = 1.

Question 3:

Which of the following sensor combinations would trigger the alert?

Options

- A) $x = (1, 1, 0)$
- B) $x = (0, 1, 1)$
- C) $x = (1, 1, 1)$
- D) $x = (1, 0, 0)$
- E) $x = (0, 0, 1)$

Correct Answer : C

Only in (1, 1, 1) the sum is 3 which is greater than 2, so the neuron fires.

Question 4:

For McCulloch-Pitts (MP) neuron model, which of the below thresholding functions describes the above scenario.

Options:

- A) $f(x) = \begin{cases} 1, & \text{if } x_1 + x_2 + x_3 > 2 \\ 0, & \text{Otherwise} \end{cases}$
- B) $f(x) = \begin{cases} 1, & \text{if } x_1 + x_2 + x_3 \geq 2 \\ 0, & \text{Otherwise} \end{cases}$
- C) $f(x) = \begin{cases} 1, & \text{if } x_1 + x_2 + x_3 > 1 \\ 0, & \text{Otherwise} \end{cases}$
- D) None of the above

Correct Answer : A

Question 5:

A team designs an MP neuron to detect when at least two out of three emergency switches are turned on. You realize the current model fires only when all three are on.

Which change will help the model fire (output 1) when at least two inputs are 1?

Options:

- A) Change the threshold from 2 to 1
- B) Change the condition from “> 2” to “ ≥ 2 ”
- C) Use weights instead of plain sums
- D) Set threshold to 3
- E) Make all inputs always 1

Correct Answer: B

Explanation:

The condition $x_1 + x_2 + x_3 > 2$ fires only when the sum is 3. To fire at 2 or more, use ≥ 2 .

Question 6:

A fault-tolerant 7-input MP neuron must output 1 only if a majority (i.e., more than 50%) of inputs are ON. The neuron receives binary signals (0/1).

Which of the following threshold values will ensure that the neuron functions as a strict majority gate?

Options:

- A. 3
- B. 4
- C. 5
- D. 6

Correct Answer: B (since majority for 7 inputs = 4 or more)

Explanation:

To ensure strict majority, the neuron must fire when ≥ 4 inputs are ON $\rightarrow$ threshold = 4.

Question 7:

Consider an MP neuron that takes 4 binary inputs with the following learning rule.

$$f(x) = \begin{cases} 1 & \text{if } x_1 + x_2 + x_3 + x_4 \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

For how many input combinations does it output 1?

- A) 8
- B) 15
- C) 1
- D) 0

Correct Answer : B

Explanation: Only when all inputs are 0, the sum is 0 $\rightarrow$ output is 0. In **15 of the 16** combinations, at least one input is 1 $\rightarrow$ output is 1.

Question 8:

A security system has 4 sensors (x_1 to x_4). An MP neuron-based controller triggers an alarm when at least 3 sensors are active. How many unique input cases will lead to an alarm?

Fill in the blank : _____

Correct Answer: 5

Explanation:

We want number of 4-bit binary combinations with $\text{sum} \geq 3$

→ Cases with 3 active inputs: $\binom{4}{3} = 4$

→ Case with all 4 active: $\binom{4}{4} = 1$

Total = 5

Question 9:

An MP neuron with 4 binary inputs gives output 1 for 11 different input combinations. What is the likelihood of θ ?

Options:

- A) 2
- B) 3
- C) 1
- 4) 4

Correct Answer : A

Explanation:

When $\theta = 2$ or 3 or 4, then the input combinations are $\binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 11$

Question 10:

You're building a neural circuit using a single perceptron that needs to identify when both security switches are on. Which Boolean function would you implement to solve this?

Options:

- A) OR

- B) XOR
- C) AND
- D) XNOR

Correct Answer: C

Explanation:

AND is linearly separable: output is 1 only for (1,1), and all others map to 0 — a line can separate this.

Question 11:

You're designing a logic gate using a perceptron. You want the gate to output 1 only when exactly one of the two inputs is 1.

Which function are you trying to implement? Can a single perceptron do it?

Options:

- A) AND, Yes
- B) XOR, No
- C) OR, Yes
- D) NOR, Yes

Correct Answer: B. XOR, No

Explanation:

XOR is not linearly separable.

Question 12:

In a smart voting system, a gate opens only if exactly one person presses a button out of two users (User A and User B).

Which logic function represents this? Can it be modeled by a perceptron?

Options:

- A) XOR, Yes
- B) XOR, No
- C) AND, Yes
- D) NOR, No

Correct Answer: B. XOR, No

Question 13: (easy)

A team is developing an AI system for identifying whether a tumor is malignant or benign based on two numerical features extracted from medical images. They decide to use a simple linear model for this binary decision task. They come across the perceptron algorithm.

Which of the following best justifies the use of the perceptron algorithm in this task?

Options:

- A) To group similar tumors into clusters
- B) To classify tumors as benign or malignant
- C) To compute the shortest path from data to prediction
- D) To solve a constrained optimization problem

Correct Answer : B

Explanation:

The perceptron is a linear classifier. It is used to assign input samples to binary classes — exactly what is needed in the tumor classification example. It is not suitable for clustering, graph problems, or optimization by itself.

Question 14: (easy)

You are designing a 2D binary classifier using the perceptron algorithm. After training the model, you decide to visualize the decision boundary and the weight vector $\vec{w}$. You notice that the line dividing the data and the vector $\vec{w}$ make an angle. Which of the following best explains the relationship between the weight vector $\vec{w}$ and the decision boundary?

Options:

- A) The weight vector is always parallel to the decision boundary
- B) The weight vector lies on the decision boundary
- C) The weight vector is perpendicular to the decision boundary
- D) The weight vector changes direction randomly with each epoch

Correct Answer : C

Data for questions 15 and 16 :

Consider you are training a perceptron algorithm on the following data, where x_1 and x_2 are features and y is a label.

x_1	x_2	y
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0	0	1
0	1	1
1	0	1
1	1	0

Question 15:

Assume that the elements in w are initialized to $[0.0, 0.0]$ and the perceptron learning algorithm is used to update the weight vector w . If the learning algorithm runs for long enough iterations then

Options

- A) The algorithm never converges
- B) The algorithm converges (i.e., no further weight updates) after some iterations
- C) The classification error remains greater than zero.
- D) The classification error becomes zero eventually

Answer : B and D

Question 16:

Assume the weights are initialized randomly in the range $[-0.5, 0.5]$, then which of the following statements are true?

Options

- A) The perceptron may fail to converge due to random initialization
- B) The perceptron will converge regardless of weight initialization
- C) Convergence time may vary depending on the initial weights
- D) Random initialization can help the model escape local minima

Correct Answer : B and C

Explanation :

For linearly separable data, the perceptron always converges regardless of the initial weights. Initial weights can affect the number of iterations needed to converge but not the fact of convergence.

The perceptron is not an optimization algorithm using gradient descent — there's no concept of local minima in this context.

Question 17:

You have a trained perceptron model, whose final weights are $w = [1, 1]$ and $b = -1$. The learning rule of the trained perceptron model is

$$y = \begin{cases} 1 & \text{if } w \cdot x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

Which of the following (x_1, x_2, y) samples would be correctly classified by this model?

Options:

- A) (1, 1, 1)
- B) (0, 0, 1)
- C) (1, 0, 0)
- D) (0, 1, 0)

Correct Answers : A, C, D

Explanation:

Question 18:

You're designing a voting system for a critical application with 7 redundant sensors. To ensure fault tolerance, the system should activate only when all sensors agree. You use an MP neuron to model this.

Which of the following threshold values will ensure true activation only when all 7 inputs are ON?

Options:

- A. 5
- B. 6
- C. 7
- D. 8

Correct Answer: C

Explanation:

To implement AND logic, the threshold must be equal to the number of inputs: all must be 1. Any lower threshold (like 6 or 5) would fire early.

Question 19:

In a voting system with 7 voters, at least 5 voters must vote YES for the system to proceed. You want to model this with an MP neuron.

What threshold value should you set?

Fill the blank_____

Correct Answer: 5

Explanation:

The neuron must fire when the sum of inputs (votes) is 5 or more, so threshold = 5.

Question 20:

Consider the following perceptron learning rule

$$y = \begin{cases} 1 & \text{if } w^T x > 0 \\ -1 & \text{otherwise} \end{cases}$$

Assume the above perceptron predicts $y = 1$, for the weight vector $w = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$, and a point $x = (a, b)$. Which of the following must be true about the relation between a and b ?

Options:

- A) $a + b < 0$
- B) $a + b = 0$
- C) $a + b > 0$
- D) $ab > 0$

Correct Answer: A

Explanation:

$$w^T x = -a - b > 0 \Rightarrow a + b < 0$$

Question 21:

Consider a perceptron with $w = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ and no bias classifies a 2D input vector using the rule: $y = \text{sign}(w^T x)$.

Which of the following points lie on the decision boundary?

Options:

- A) (2, 2)
- B) (1, -1)
- C) (3, -3)
- D) (0.5, 0.5)

Correct Answer: B and C

Explanation:

$w^T x = -x - y = 0 \Rightarrow x + y = 0$ defines the decision boundary.

Question 22:

Given a perceptron with weight vector $[2, 3, -1]$ and bias -2 , what will be the output when the input vector is $[1, 0, -1]$?

Fill the answer : _____

Correct Answer : 1

Questions 23 to 25

Instructions for the following questions

Assume perceptron learning without bias and use the following update rules:

If the example is from class $y = 1$ and $w \cdot x < 0$, then $w = w + x$

If the example is from class $y = 0$ and $w \cdot x \geq 0$, then $w = w - x$

All feature vectors have two dimensions. Pick the examples in the order mentioned in the table.

Question 23:

The perceptron is initialized with weight vector $w = (3,3)$. Consider the following training dataset.

x_1	x_2	y
2	2	1
-1	-2	0
-2	-1	0

How many updates will the perceptron perform in one full pass over the data?

Fill the blank : _____

Correct answer is : 0

Question 24:

The dataset is

x_1	x_2	y
1	0	1
0	1	1
-1	-1	0

Assume perceptron starts with $w = (0, 0)$, and training continues until convergence.

Which of the following could be a valid final weight vector after convergence?

- (A) (1, 0)
- (B) (-1, -1)
- (C) (0, 1)
- (D) (1, 1)

Correct Answer: D

Explanation: for $w = (1, 1)$ – all the inputs are classified correctly.

Question 25:

The dataset is

x_1	x_2	y
-1	-2	1
1	1	0

Assume perceptron learning starts with $w = (0, 0)$. The algorithm continues until convergence.

What is the final weight vector after convergence?

- A) (1,2)
- B) (-2, -1)
- C) (-1,-1)
- D) None of these.

Correct Answer: C

Explanation: for $w = (-1, -1)$ – all the inputs are classified correctly.

Question 26:

Which of the following points best explains why adding a bias term in a perceptron model helps in classification?

Options:

- A) It shifts the decision boundary away from the origin
- B) It rotates the decision boundary
- C) It increases the number of epochs required for convergence
- D) None of the above

Correct Answer : A

Question 27:

Which of the following conditions must be true for the perceptron learning algorithm to guarantee convergence?

Options:

- A) The data must be linearly separable
- B) The weight vector must be initialized to non-zero values
- C) The number of data points must be finite
- D) The algorithm must include a bias term

Correct Answers: A, C

Explanation: The perceptron guarantees convergence only if the data is finite and linearly separable.

Bias is helpful but not necessary for convergence; initialization does not affect convergence guarantee.

Question 28:

What does it mean for two sets P and N to be absolutely linearly separable in n-dimensional space?

Options:

- A) There exists a hyperplane that correctly classifies all points from both sets
- B) Every point from P lies on the same coordinate axis
- C) Each set must have an equal number of points
- D) The points in P are closer to the origin than those in N

Correct Answer: (A)

Explanation : Linear separability means there exists a set of weights $w_0, w_1, \dots, w_n$ that separates the two sets with a hyperplane.